

Week 5 – User Management, Authentication, and Access Control

The purpose of week 5 is to strengthen the security posture of the Ubuntu server by implementing proper user management, permissions, and access control.

This includes:

- Creating new users and assigning appropriate roles
- Managing system groups to enforce least-privilege principles
- Controlling elevated access through sudo
- Configuring secure file permissions
- Ensuring that authentication and access rules comply with best security practices

This forms the foundation for a secure multi-user Linux environment and aligns with industry standards for server administration.

Linux Mint uses SSH access to remotely manage users on the Ubuntu server.

I added a user named studentuser to the server:

```
katana@katana-VirtualBox:~$ sudo adduser studentuser
[sudo] password for katana:
Sorry, try again.
[sudo] password for katana:
info: Adding user `studentuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `studentuser' (1001) ...
info: Adding new user `studentuser' (1001) with group `studentuser (1001)' ...
info: Creating home directory `/home/studentuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for studentuser
Enter the new value, or press ENTER for the default
  Full Name []: kurt katana
  Room Number []: 12
  Work Phone []: 07440774125
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `studentuser' to supplemental / extra groups `users' ...
info: Adding user `studentuser' to group `users' ...
katana@katana-VirtualBox:~$
```

To allow controlled elevated privileges, studentuser was added to a dedicated group:

```
katana@katana-VirtualBox:~$ sudo groupadd adminusers
katana@katana-VirtualBox:~$ sudo usermod -aG adminusers studentuser
katana@katana-VirtualBox:~$ groups studentuser
studentuser : studentuser users adminusers
katana@katana-VirtualBox:~$
```

I added studentuser to adminusers group and devteam. After that, I granted it full sudo rights:

```
# Host alias specification
%adminusers ALL=(ALL:ALL) ALL
# User alias specification

# Cmnd alias specification

# User privilege specification
```

To verify the sudo access I logged in as the user:

```
katana@katana-VirtualBox:~$ su - studentuser
password:
studentuser@katana-VirtualBox:~$ sudo ls /root
[sudo] password for studentuser:
studentuser@katana-VirtualBox:~$
```

I created a directory for the devteam group:

```
studentuser@katana-VirtualBox:~$ sudo mkdir /srv/dev
studentuser@katana-VirtualBox:~$ sudo chgrp devteam /srv/dev
studentuser@katana-VirtualBox:~$ sudo chmod 770 /srv/dev
studentuser@katana-VirtualBox:~$
```

I also restricted SSH access to only approved users:

```
# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
AllowUsers vboxuser studentuser
```